Student Result Management System

SOURCE CODE DOCUMENTATION

Program Description

This document details the refined C++ implementation of an interactive console application designed for processing student academic scores. The software dynamically accepts student metadata alongside marks scored across three independent modules, computes cumulative and mean metrics, and evaluates standard pass/fail threshold fulfillment.

C++ Source Code

```cpp
#include <iostream>
#include <string>

void displayResult(std::string name, std::string rollNumber, float subject1, float subject2, float s
    float total = subject1 + subject2 + subject3;
    float average = total / 3.0;

    std::cout << "\n--- Student Report ---" << std::endl;
    std::cout << "Roll Number: " << rollNumber << std::endl;
    std::cout << "Name: " << name << std::endl;
    std::cout << "Total Marks: " << total << std::endl;
    std::cout << "Average Mark: " << average << std::endl;

    if (average >= 50.0) {
        std::cout << "Status: PASSED" << std::endl;
    } else {
        std::cout << "Status: FAILED" << std::endl;
    }
    std::cout << "----------------------\n" << std::endl;
}

int main() {
    std::string name;
    std::string rollNumber;
    char choice;

    do {
        std::cout << "Enter Student Roll Number: ";
        std::cin >> rollNumber;

        std::cin.ignore();
        std::cout << "Enter Student Name: ";
        std::getline(std::cin, name);

        float sub1, sub2, sub3;
        std::cout << "Enter marks for Subject 1: ";
        std::cin >> sub1;
        std::cout << "Enter marks for Subject 2: ";
        std::cin >> sub2;
```

```
        std::cout << "Enter marks for Subject 3: ";
        std::cin >> sub3;

        displayResult(name, rollNumber, sub1, sub2, sub3);

        std::cout << "Do you want to enter marks for another student? (y/n): ";
        std::cin >> choice;
        std::cout << std::endl;

    } while (choice == 'y' || choice == 'Y');

    std::cout << "Program exited. Thank you!" << std::endl;
    return 0;
}
```